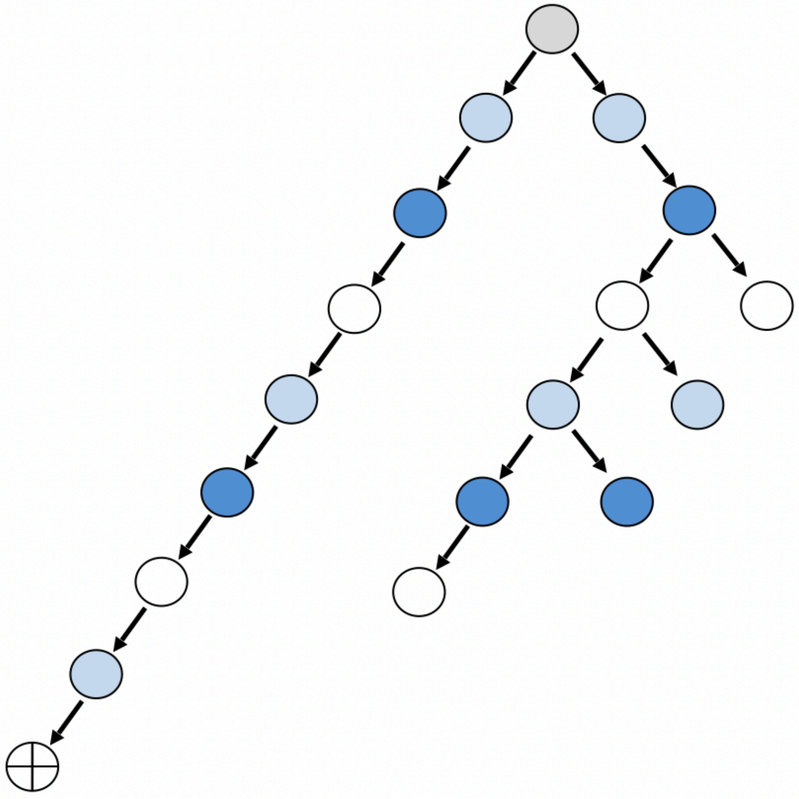
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Robot Motion & Planning

Lecture 09B - Graph Search



Agenda

1. Informed Search
2. Best First Search
3. Greedy Search
4. A* Search

Agenda

1. **Informed Search**
2. Best First Search
3. Greedy Search
4. A* Search

- **In uninformed searches**, we saw that the time complexities were exponential.

- **Informed search** tries to use information to reduce the time required for search.

- **The informed search relies on additional knowledge about the problem or domain**
  - This knowledge is frequently expressed through heuristics or “rules of thumb”
  - This knowledge is used to distinguish more promising paths towards a goal
  - This knowledge may mislead the search, depending on the quality of the heuristic

- In general, performs much better than uninformed search!

Agenda

1. Informed Search
2. **Best First Search**
3. Greedy Search
4. A* Search

- **Best First Search relies on an evaluation function**

  - The evaluation function gives an indication of how useful it would be to expand a node
  - Best First Search is a family of search methods with various evaluation functions
  - It often gives an estimate of the distance to the goal

- **Define the evaluation function $f(n)$ for node n**

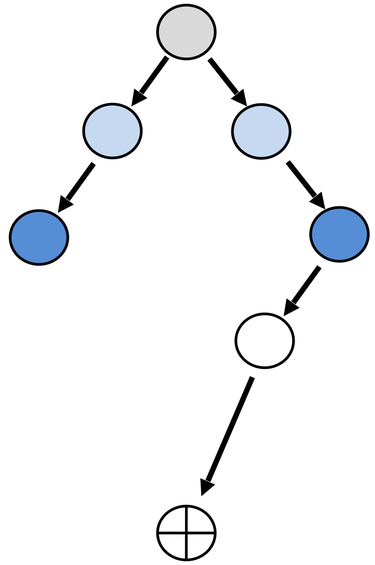
  - This function returns a number representing the desirability (or lack thereof) of expanding node n.
  - When nodes are ordered for expansion, such nodes with the best evaluation function are expanded first,
    - We say this implements a **Best First Search** strategy.

- **In Best First Search, the node with the lowest (or highest) value is expanded first**
  - The name is a little misleading: the node with the lowest value for the evaluation function is not necessarily one that is on an optimal path to a goal
  - If we really knew which one was the best, there's no need to do a search

Agenda

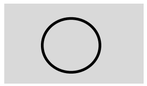
1. Informed Search
2. Best First Search
3. **Greedy Search**
4. A* Search

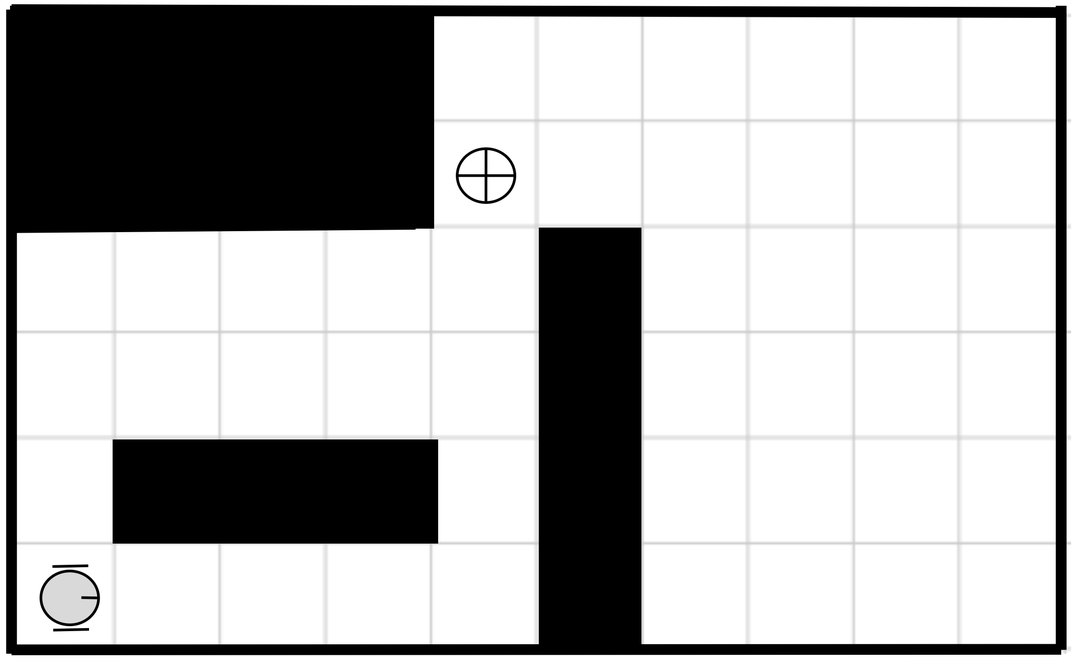
- **Expand the node that has the shortest estimate of cost to goal.**
  - $f(n) = h(n) = d(n,g)$
  - Where $d(n,g)$ is Euclidean distance from node n to the goal g



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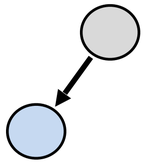
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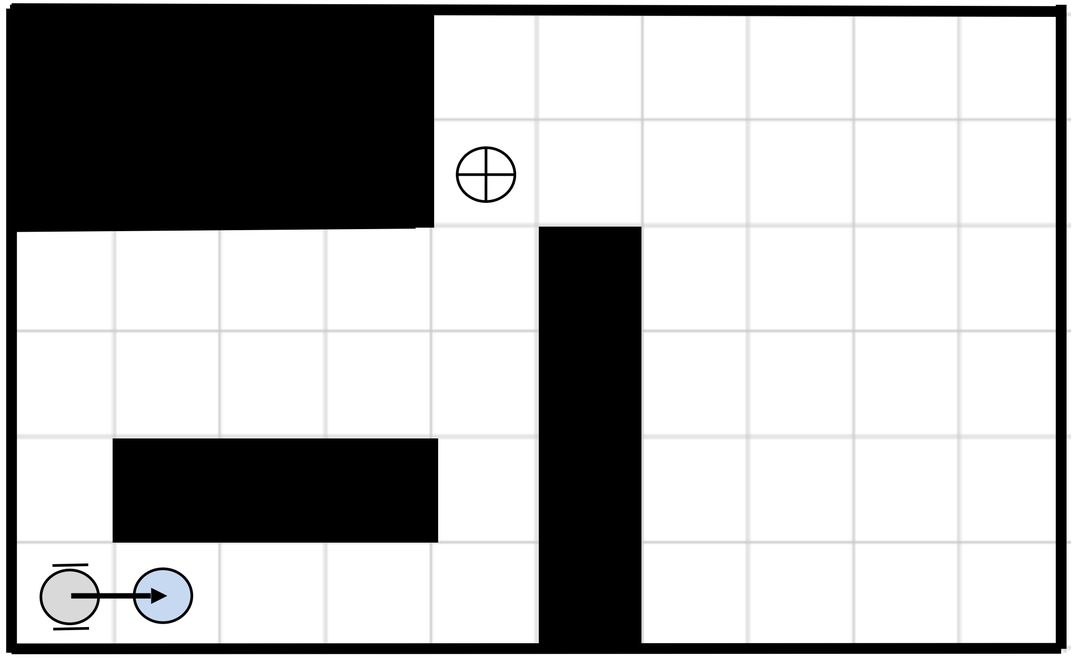


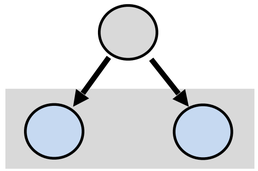


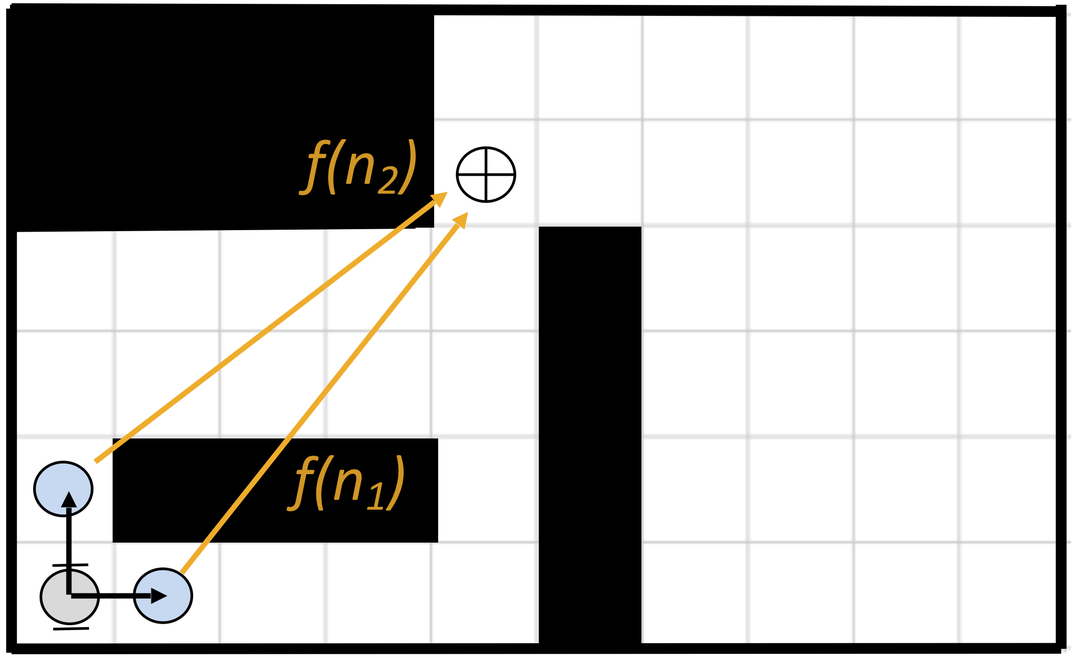
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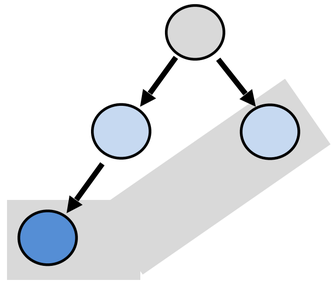
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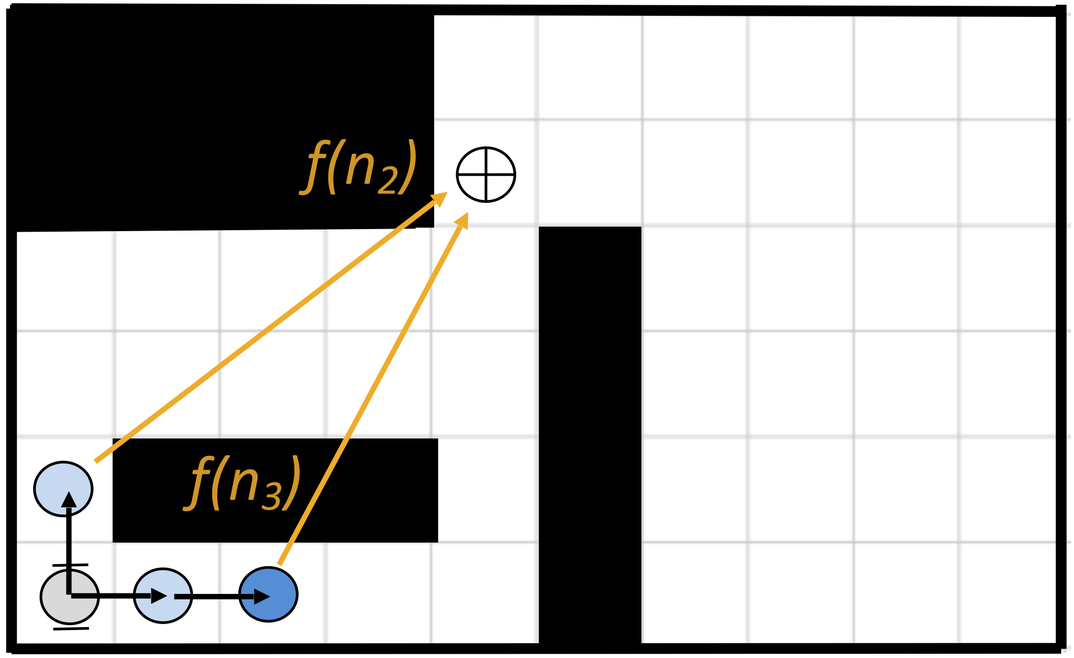


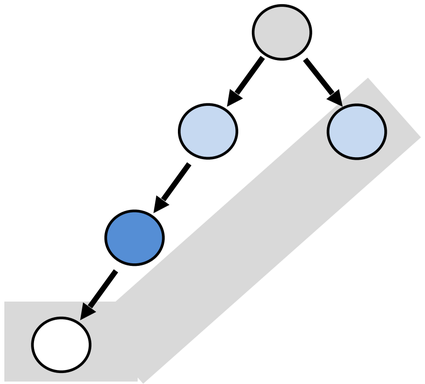


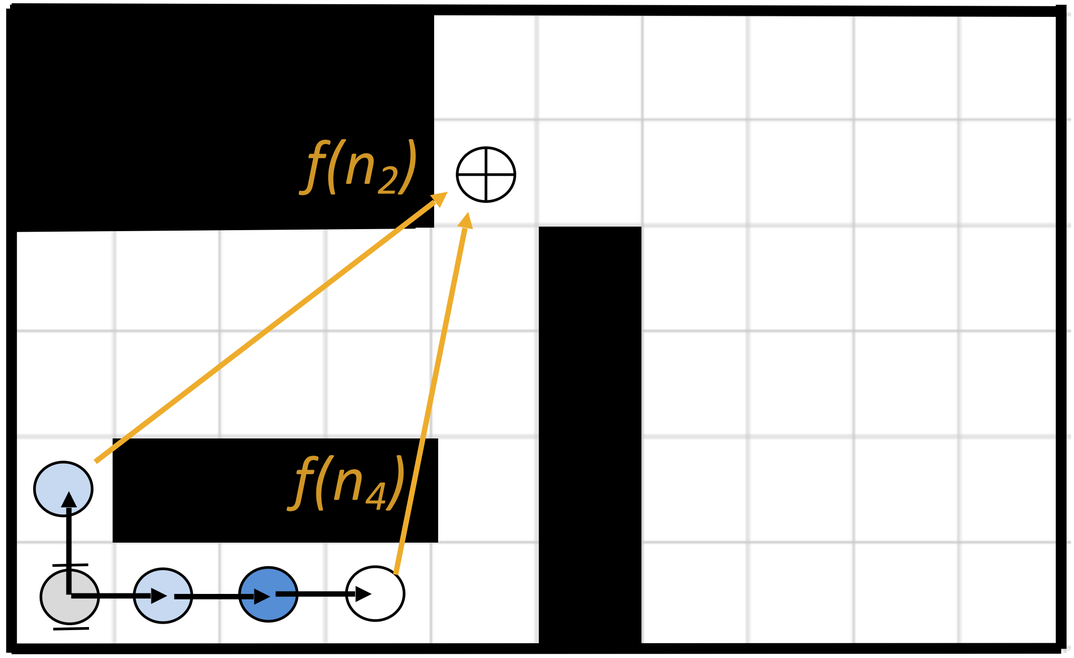


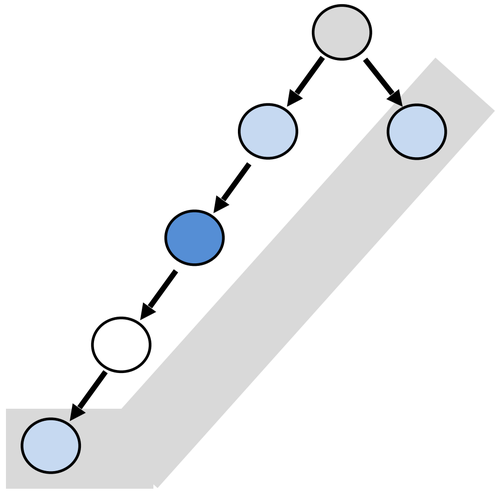

$f(n_2)$
$f(n_1)$

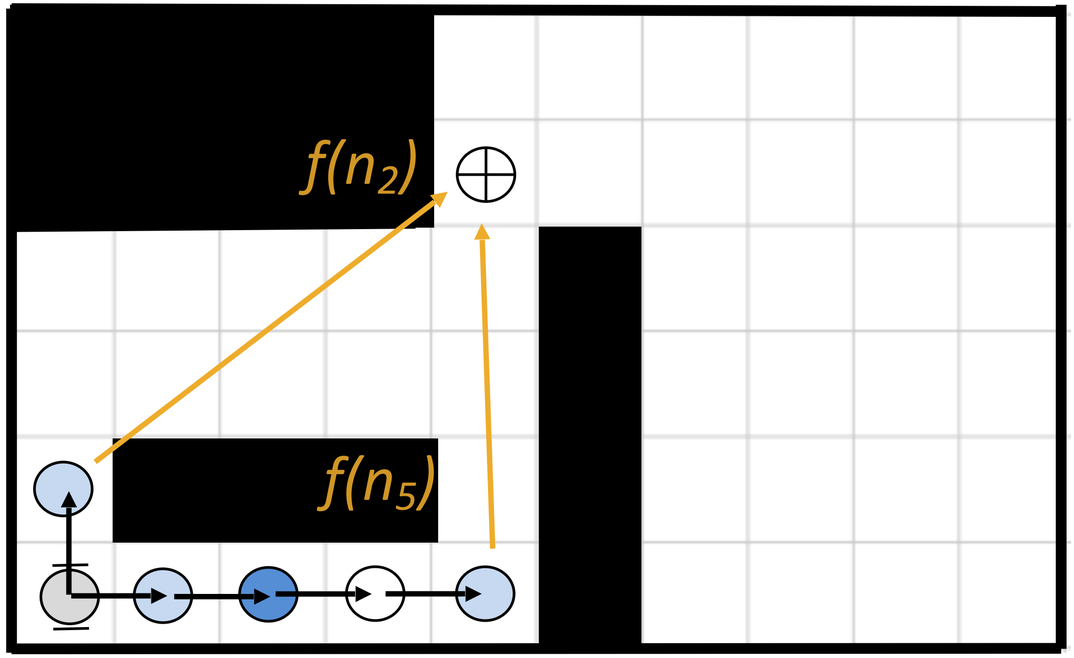



$f(n_2)$
$f(n_3)$



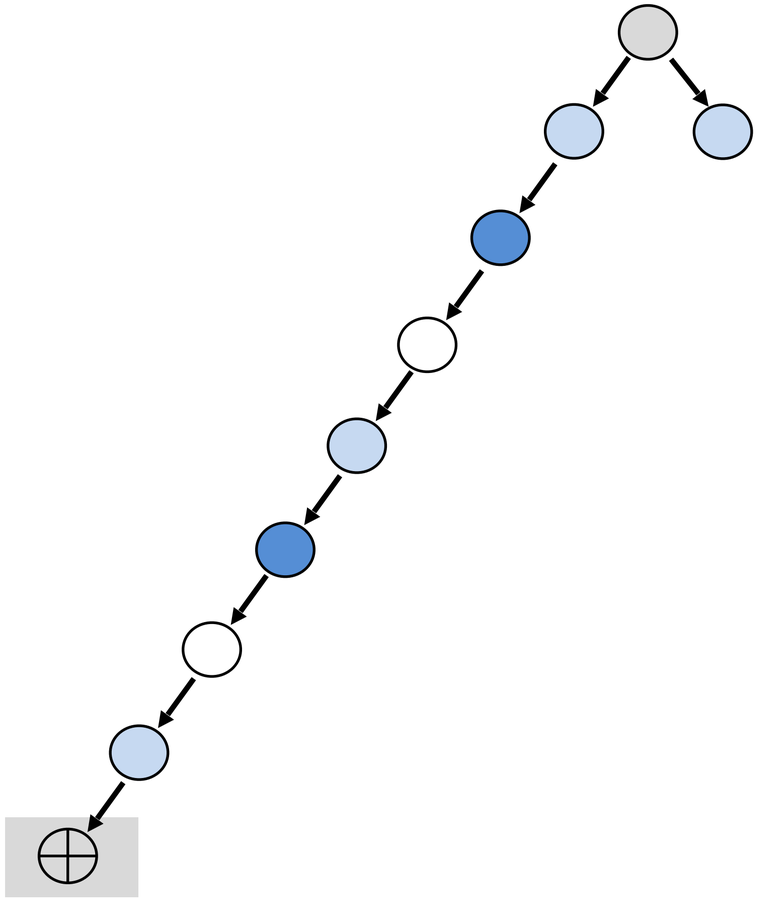

$f(n_2)$
$f(n_4)$

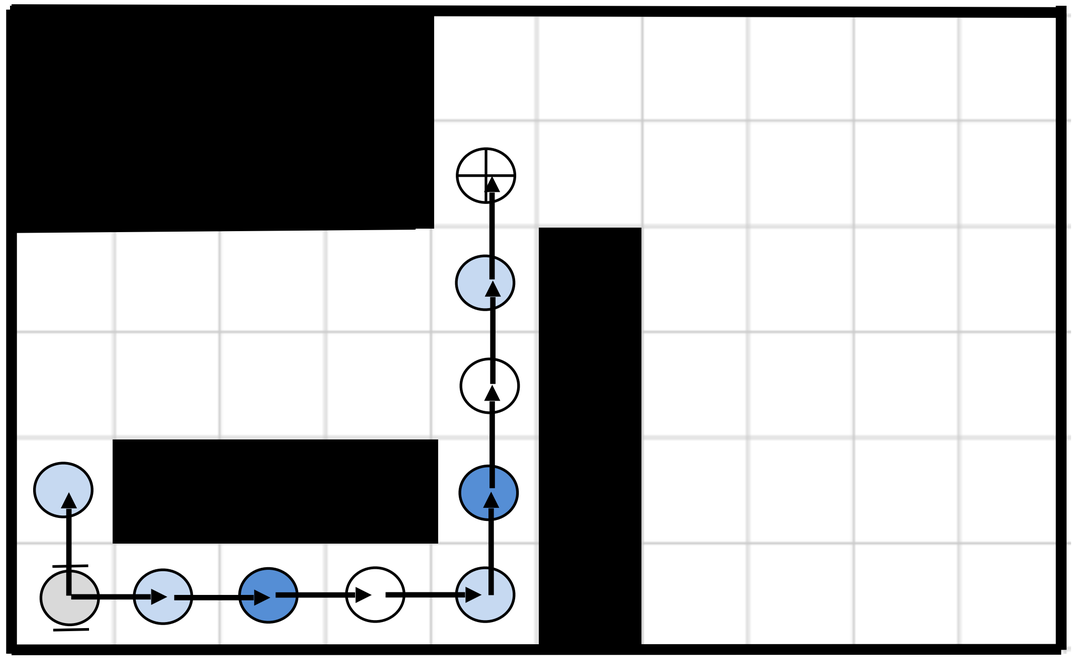



$f(n_2)$
$f(n_5)$

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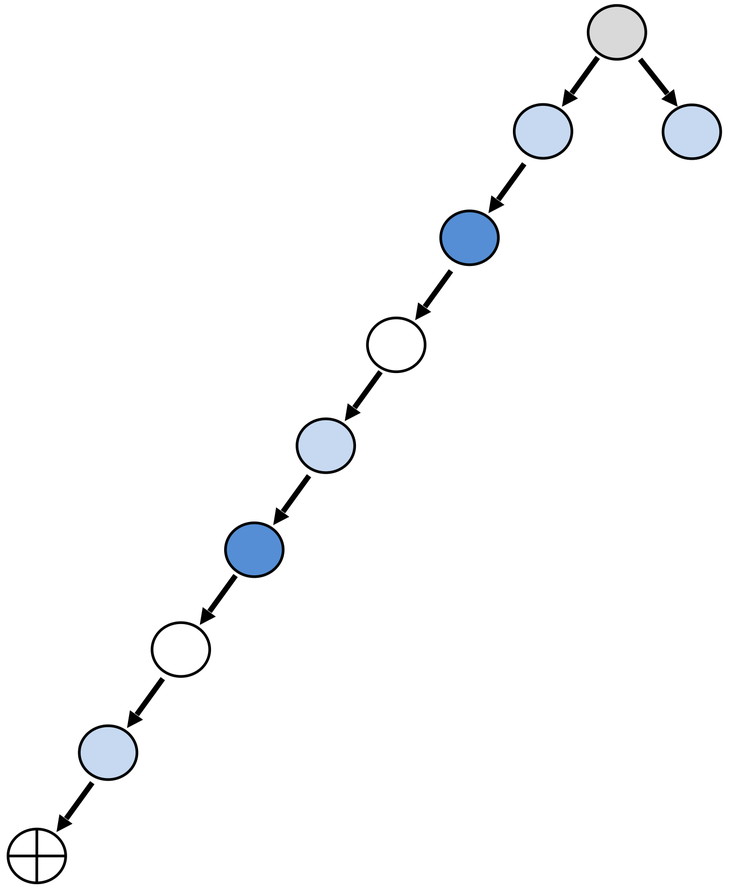
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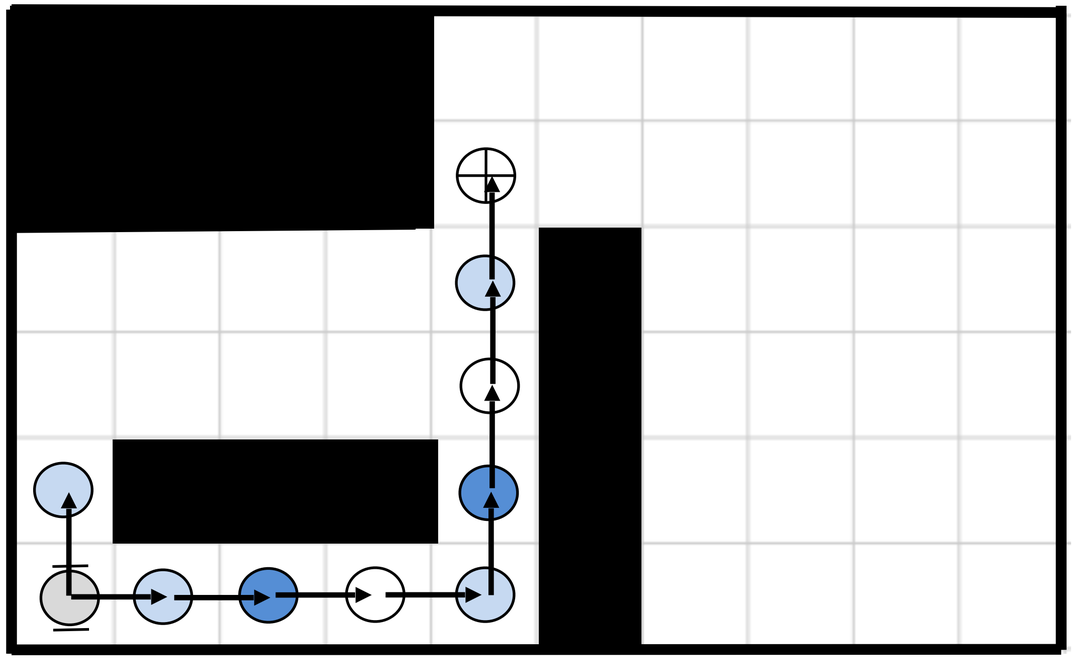




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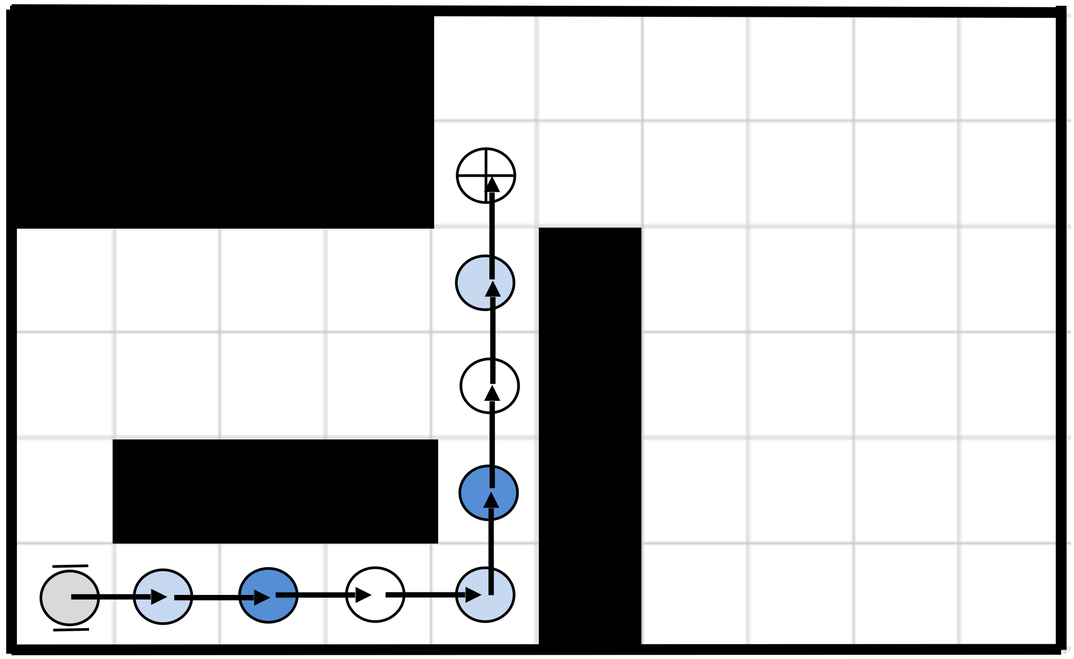
Robot Motion & Planning





- **Characteristics**

  - The algorithm is complete since it systematically checks all possible options.
  - The traj will be sub-optimal, but fast to calculate in many workspaces with convex obstacles.



Agenda

1. Informed Search
2. Best First Search
3. Greedy Search
4. **A* Search**

- **A* Search**

  - Combines greedy and uniform-cost search to find the (estimated) cheapest path through the current node
    - Uniform cost search uses information about path cost from start to fringe nodes
    - Greedy search uses information about path cost from fringe nodes to the goal node

- The evaluator function for A* search combines functions from greedy and uniform-cost

$$f(n) = g(n) + h(n)$$

where $g(n)$ = cost from start to n

$h(n)$ = estimated cheapest cost from n to goal

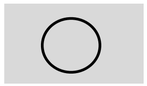
- **The evaluator function $f(n)$ works well, if the function $h(n)$ is an admissible heuristic**

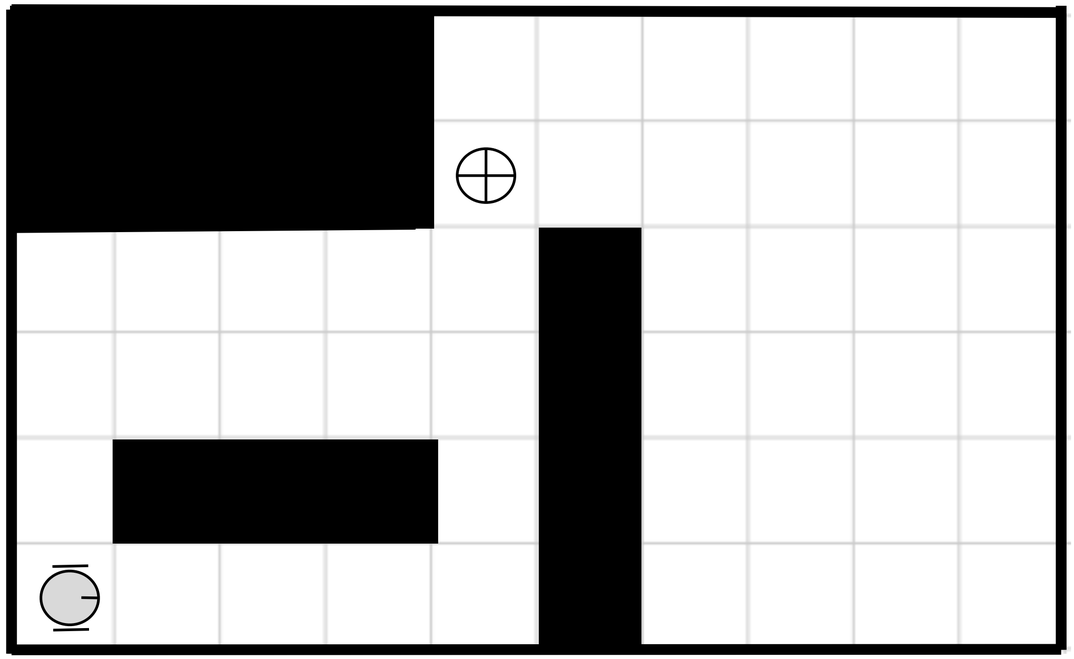
  - Def'n: An admissible heuristic is one that never overestimates the cost to reach the goal

  - Example: The straight line distance from n to the goal

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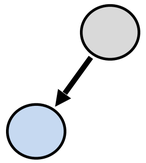
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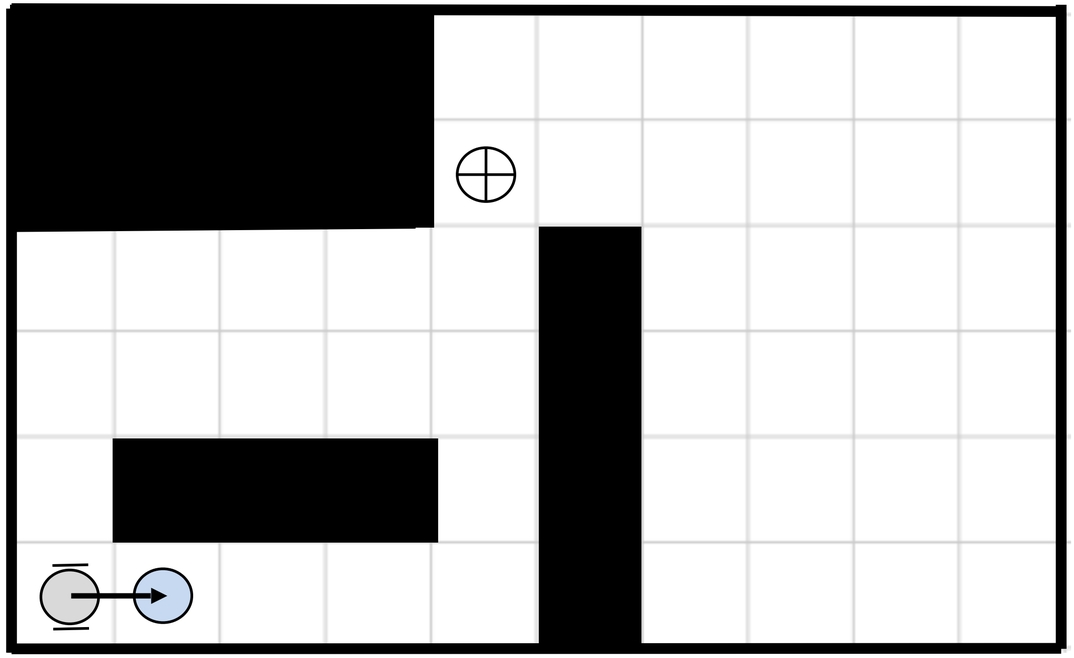


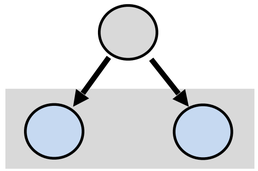


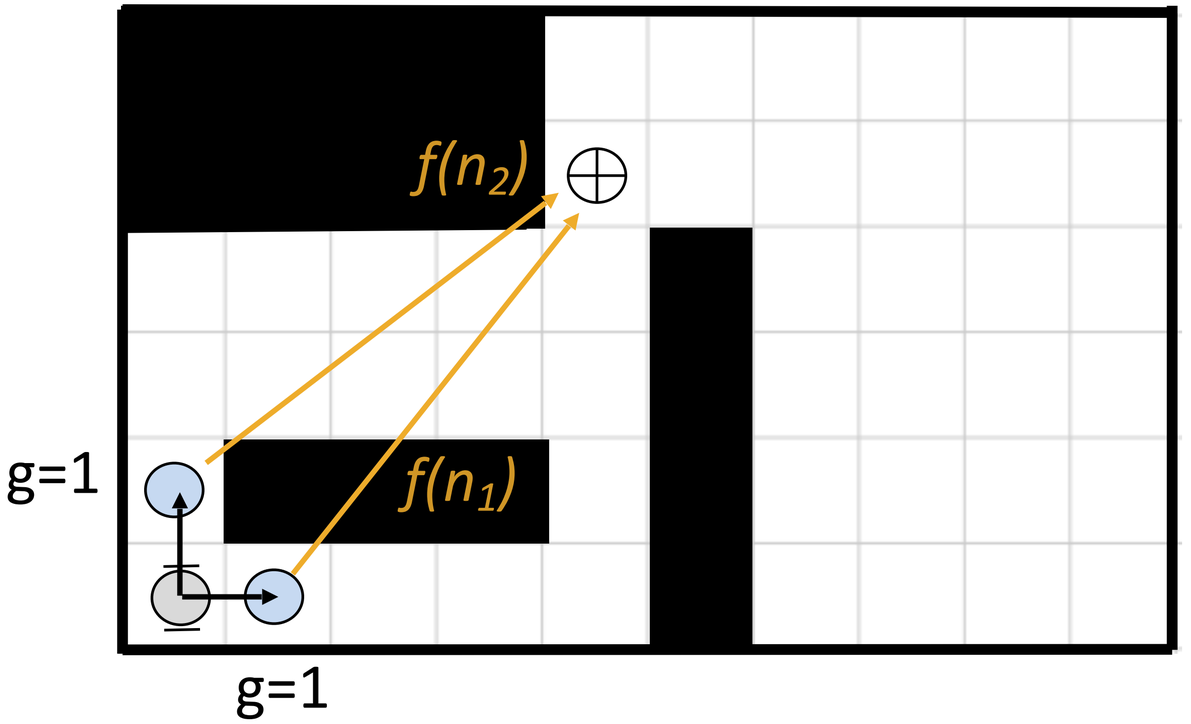
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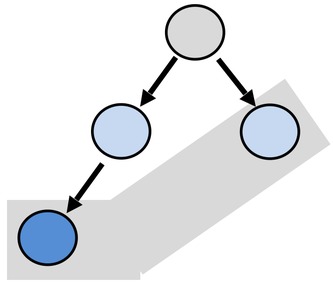
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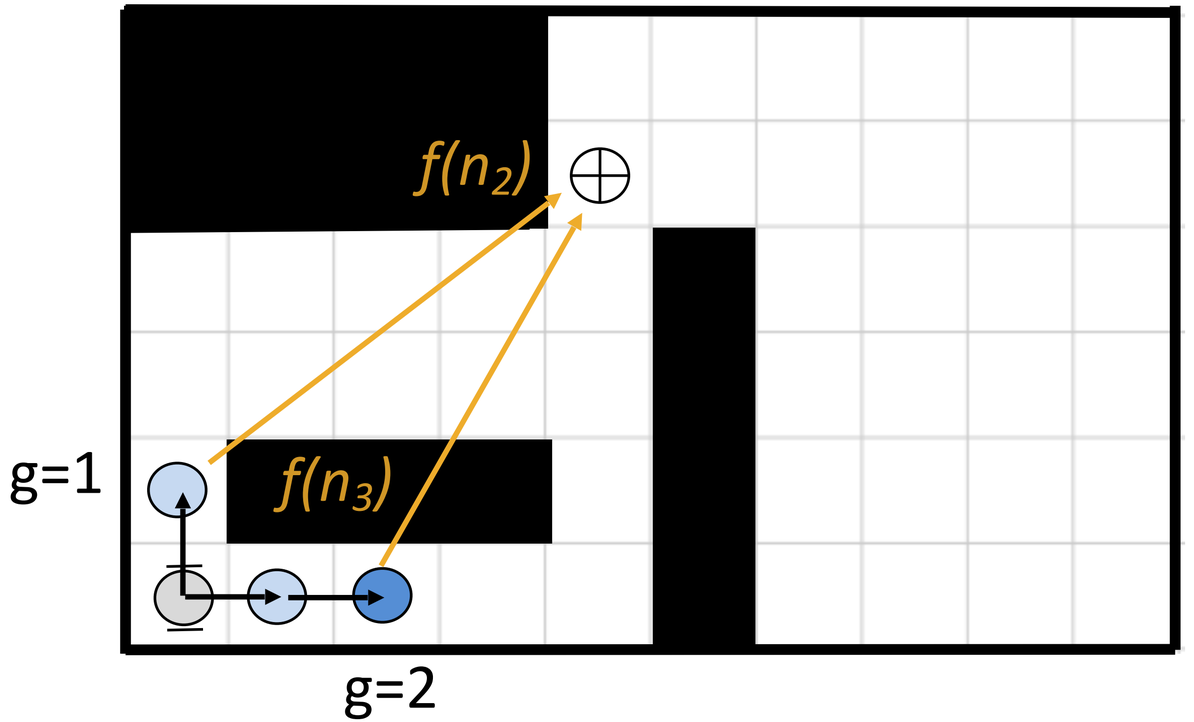


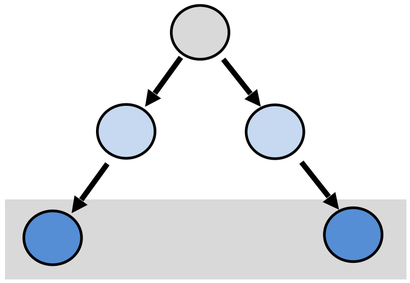


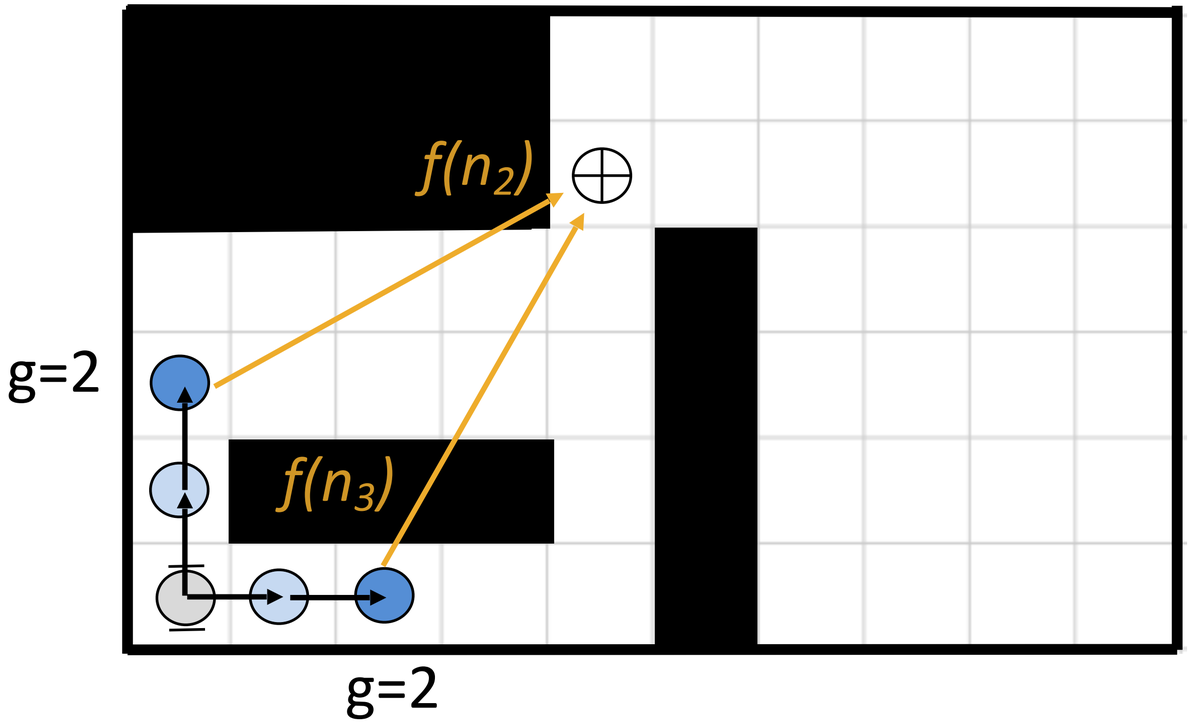


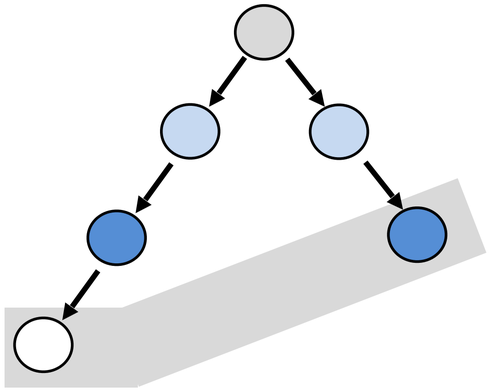

$f(n_2)$
$f(n_1)$
g=1
g=1

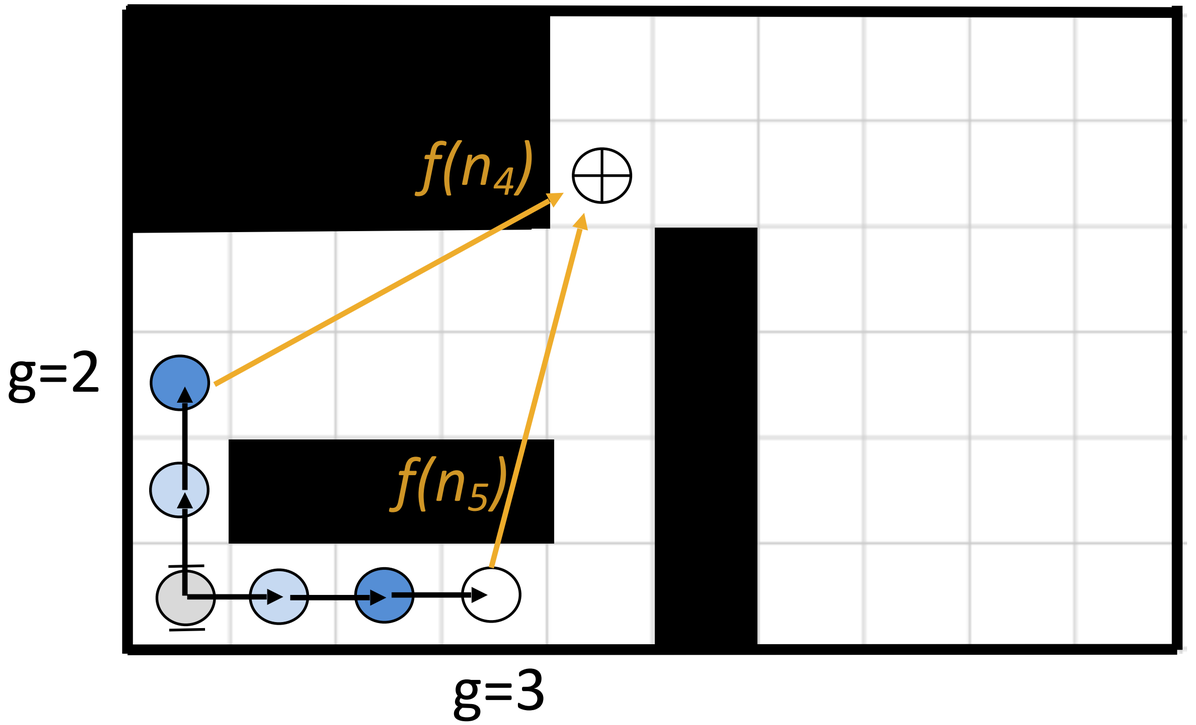



$f(n_2)$
$f(n_3)$
g=1
g=2



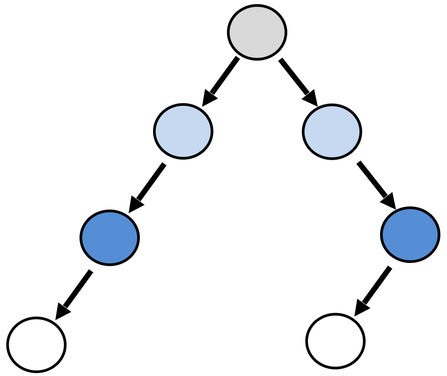

$f(n_2)$
g=2
$f(n_3)$
g=2

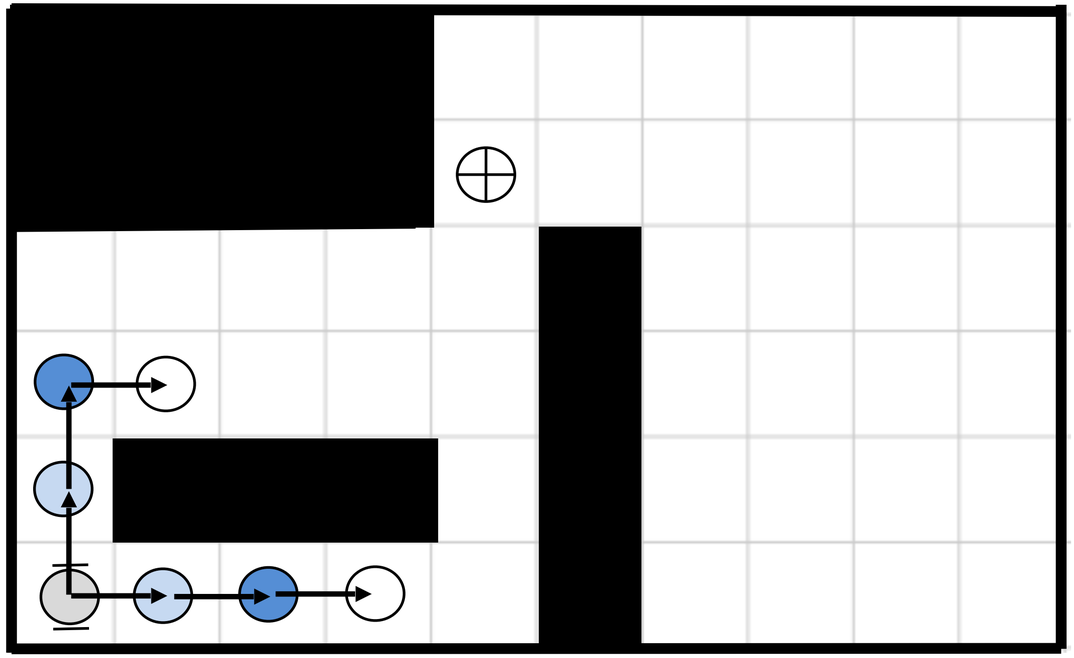


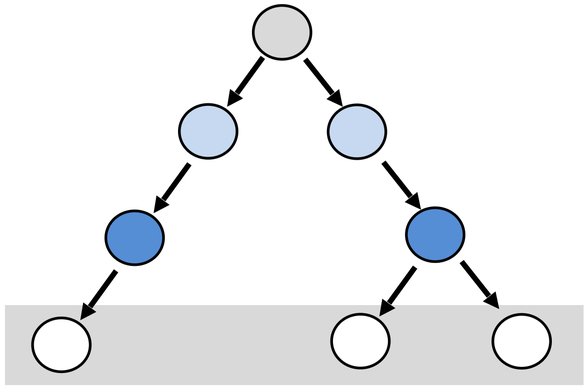

$f(n_4)$
$f(n_5)$
g=2
g=3

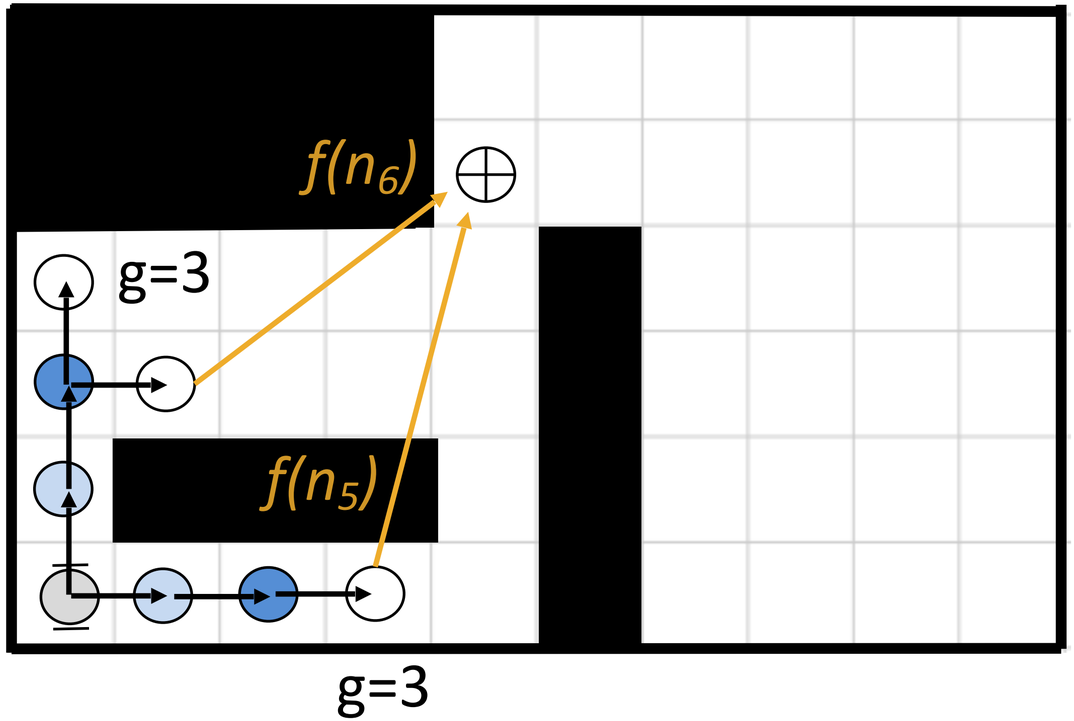
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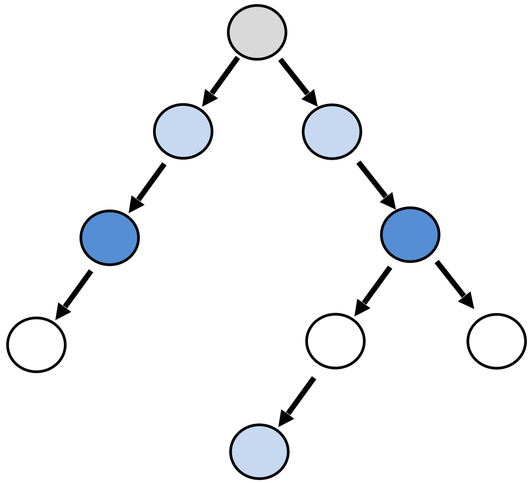


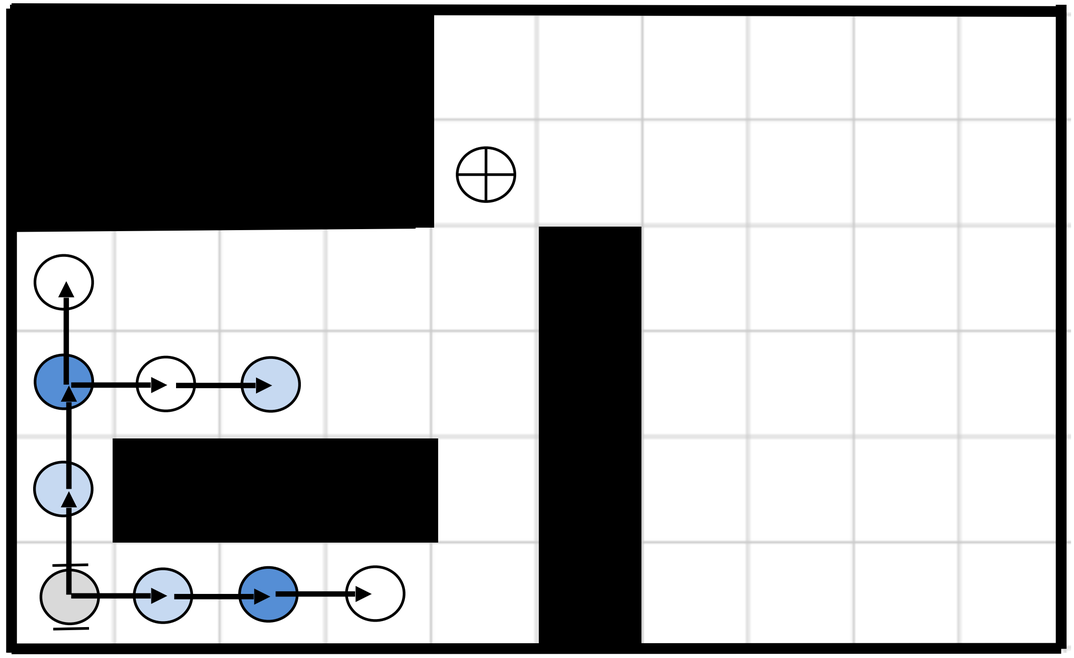


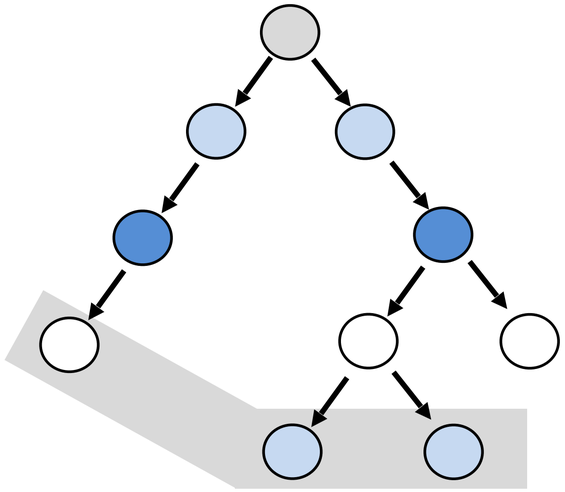

$f(n_6)$
g=3
$f(n_5)$
g=3

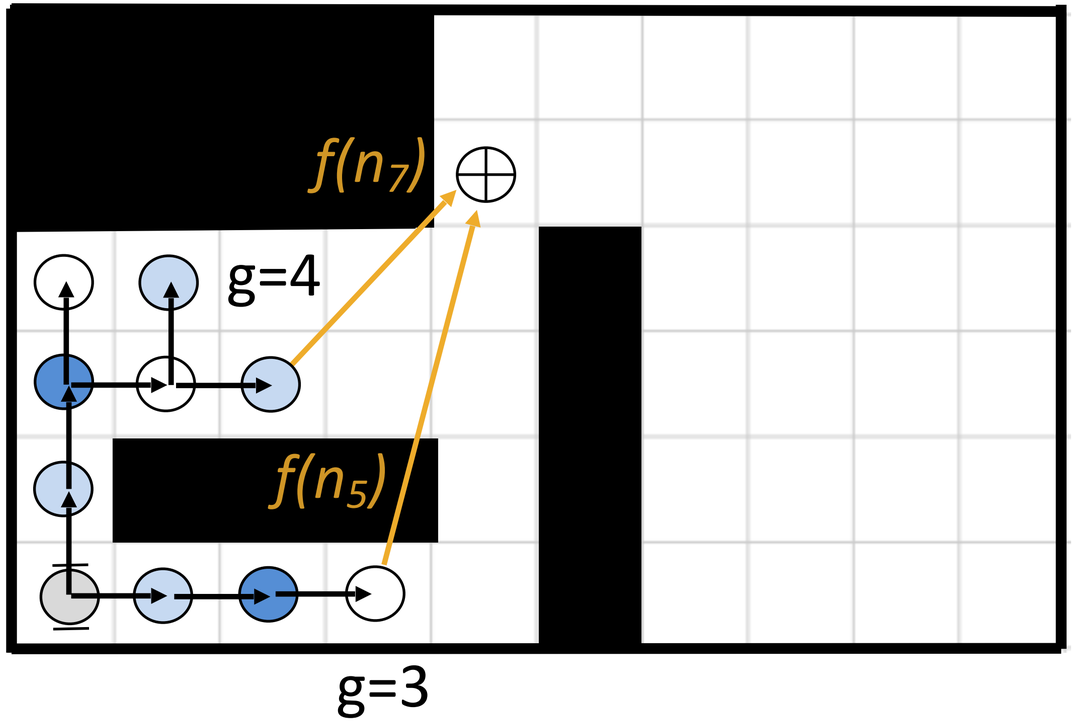
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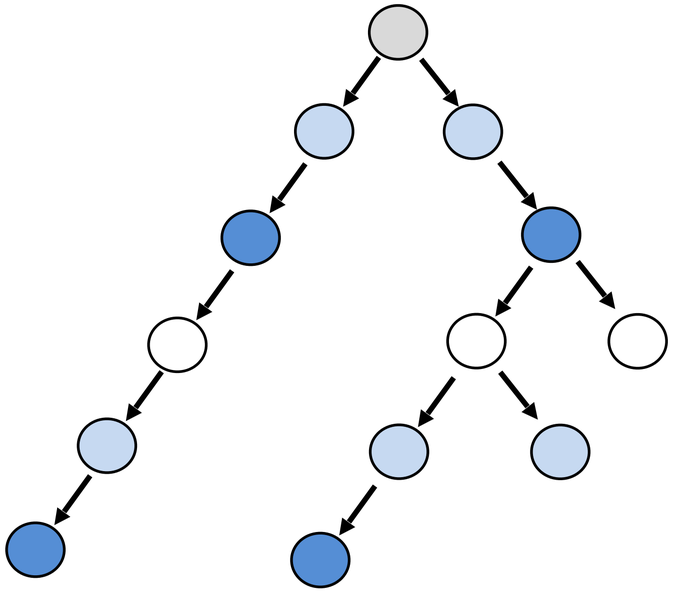


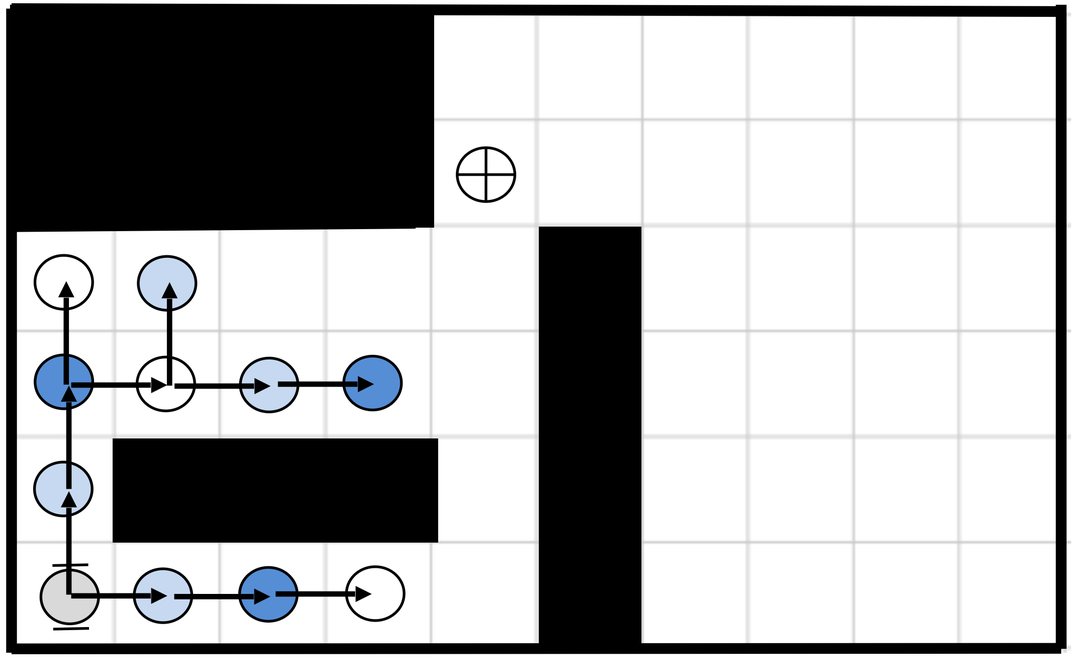


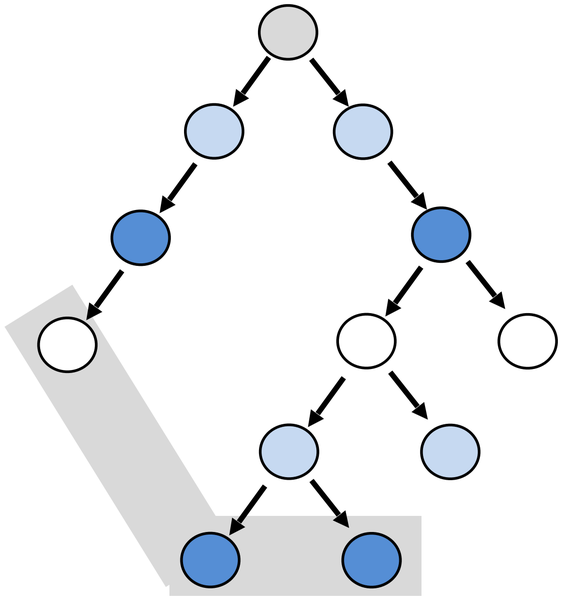

$f(n_7)$
g=4
$f(n_5)$
g=3

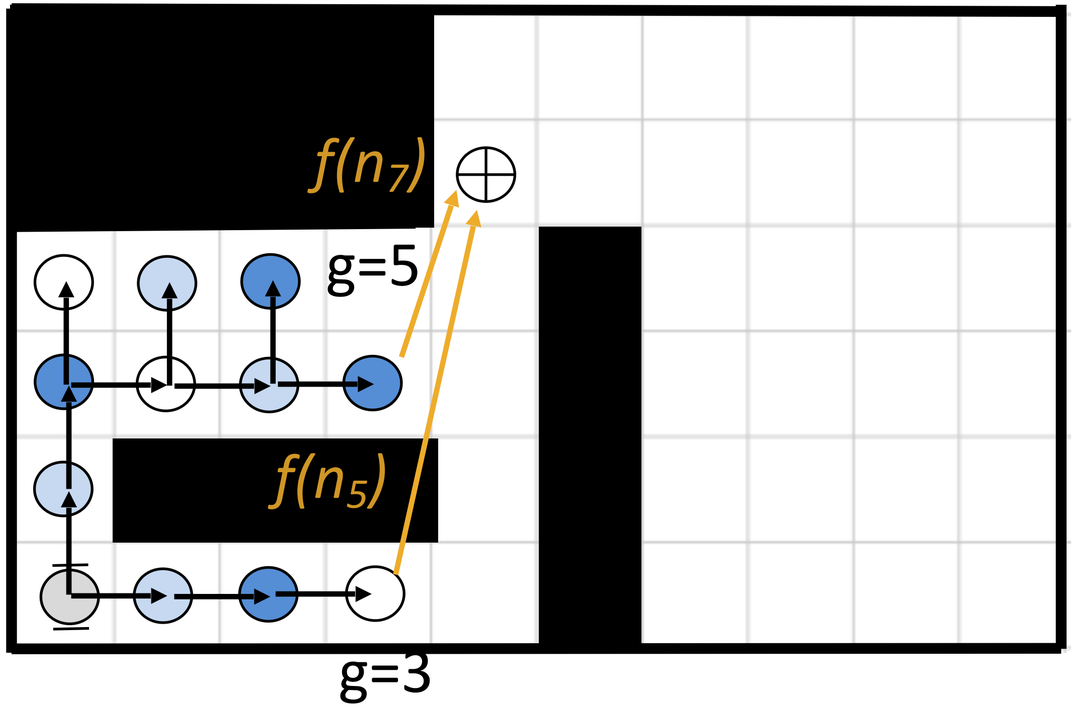
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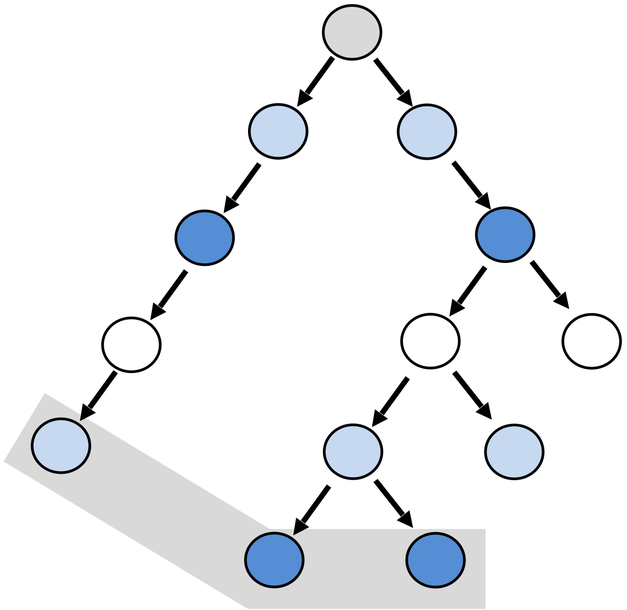
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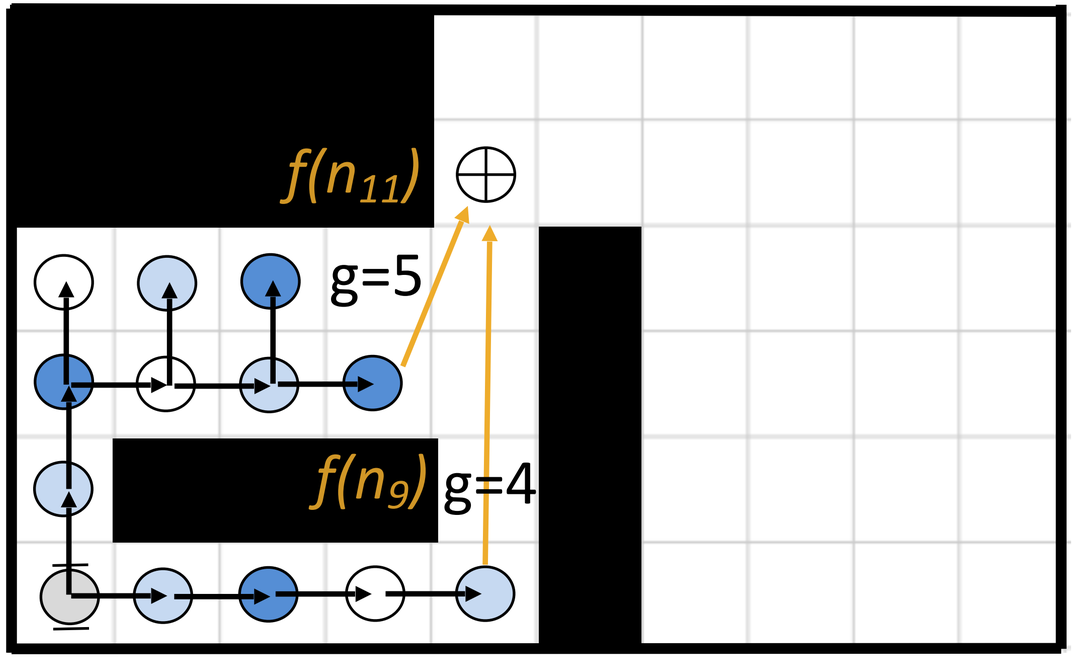


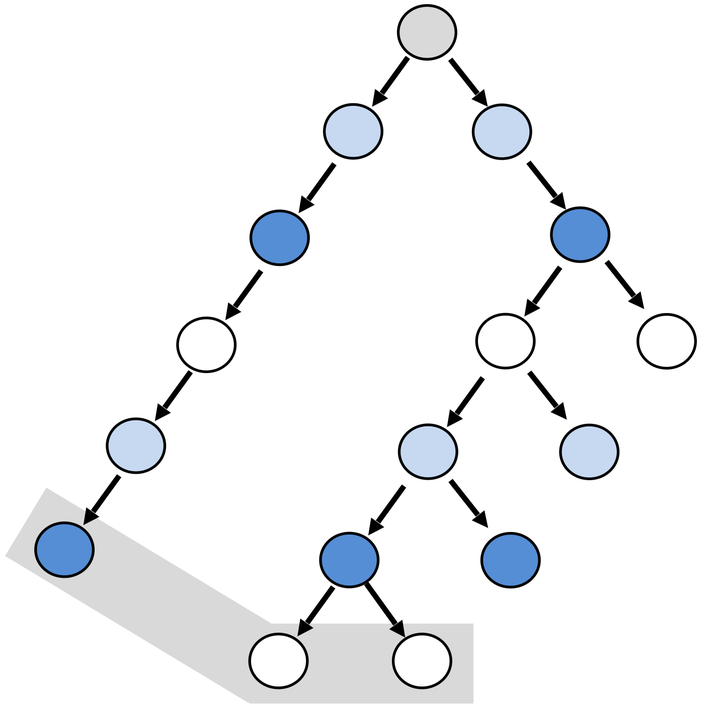


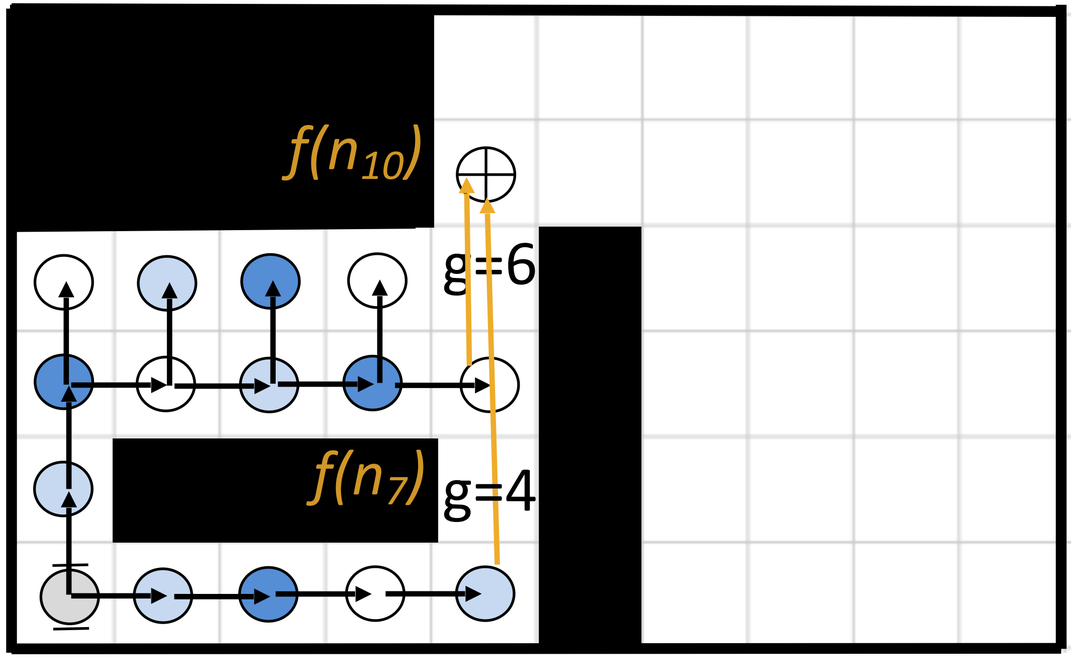



$f(n_7)$
g=5
$f(n_5)$
g=3



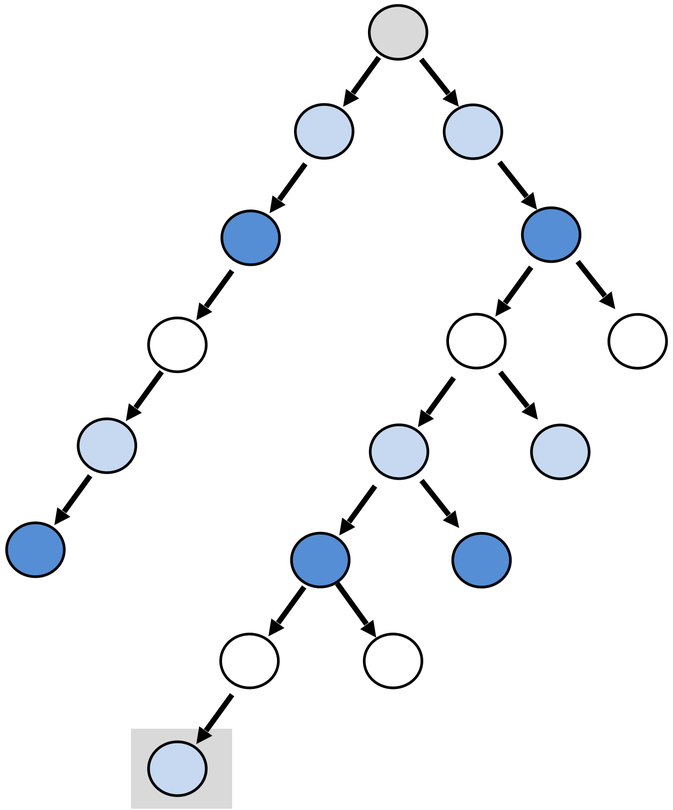

$f(n_{11})$
g=5
$f(n_9)$
g=4

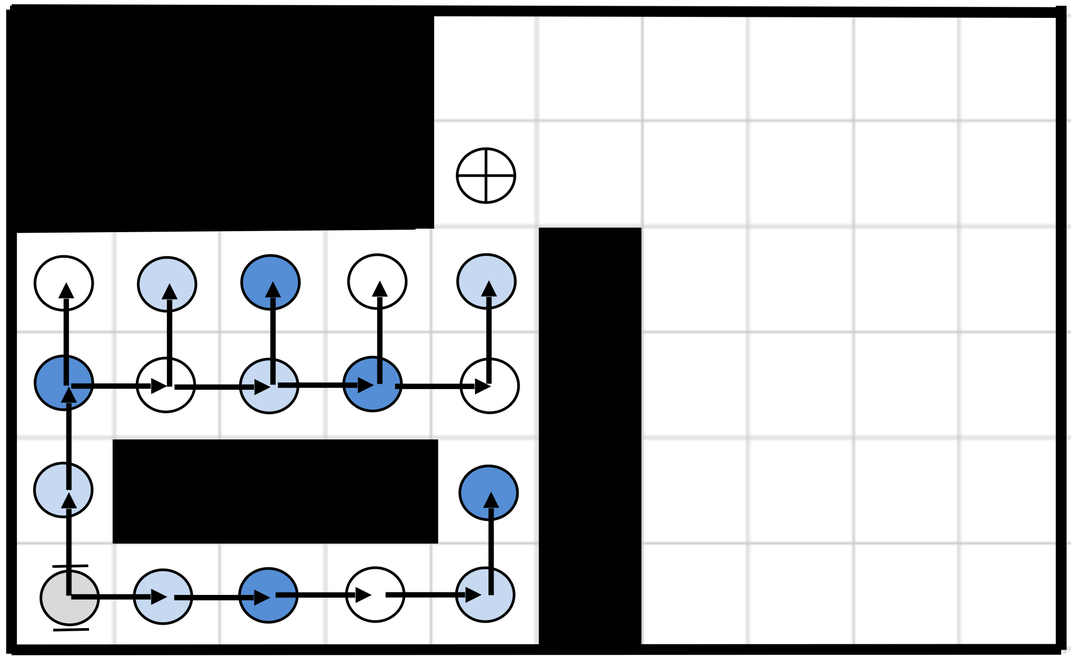



$f(n_{10})$
g=6
$f(n_7)$
g=4

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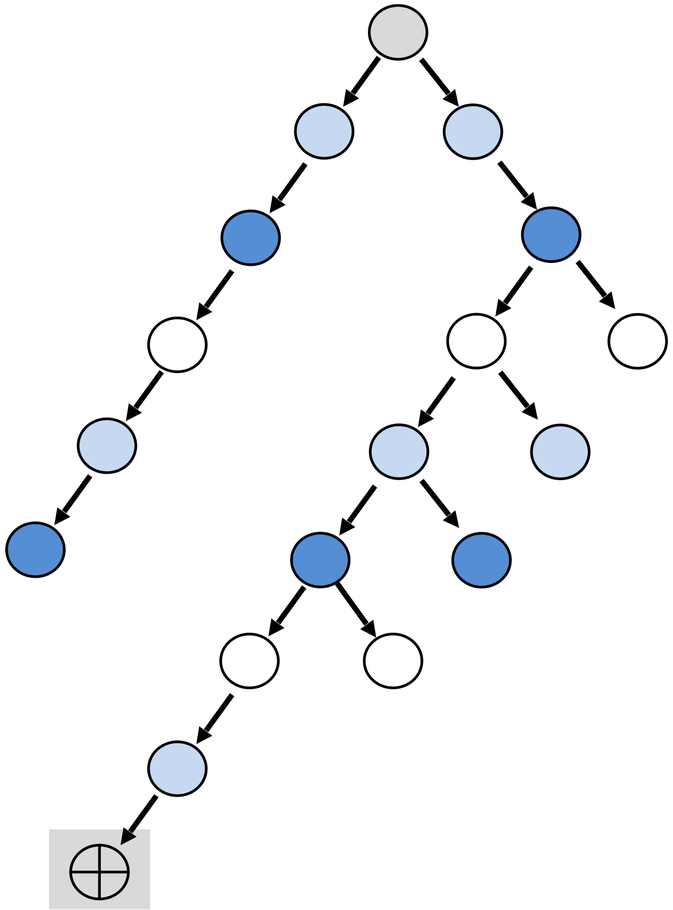
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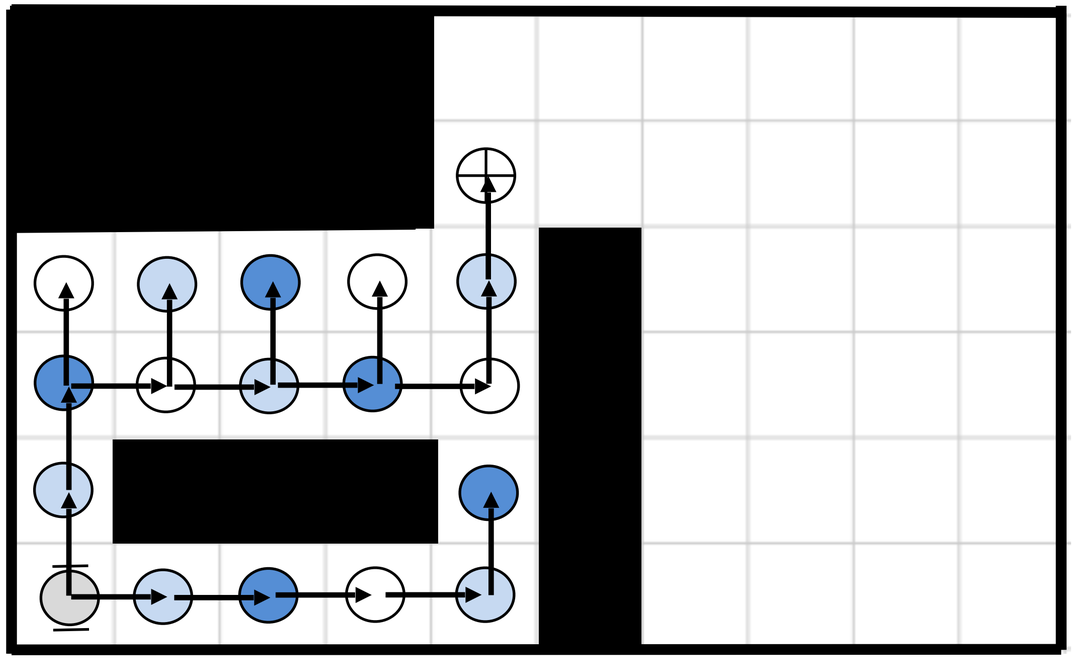




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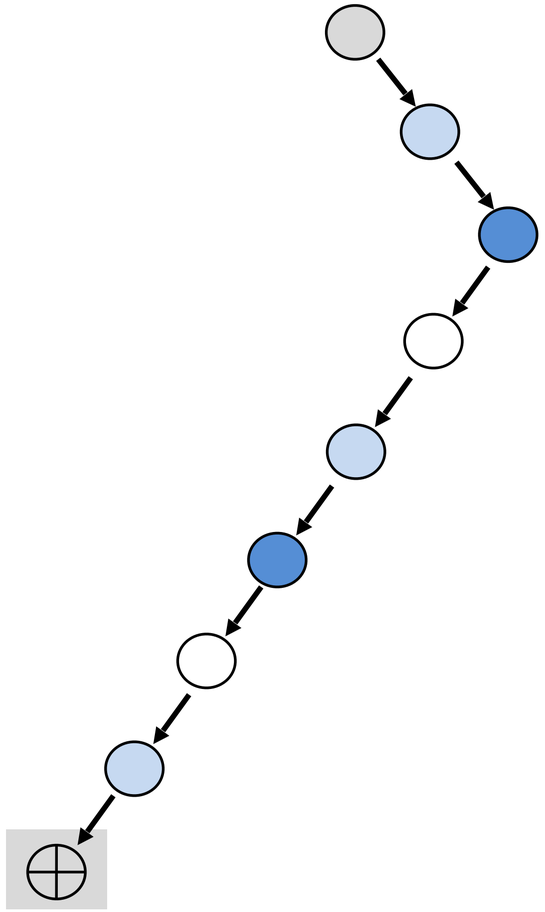
Robot Motion & Planning

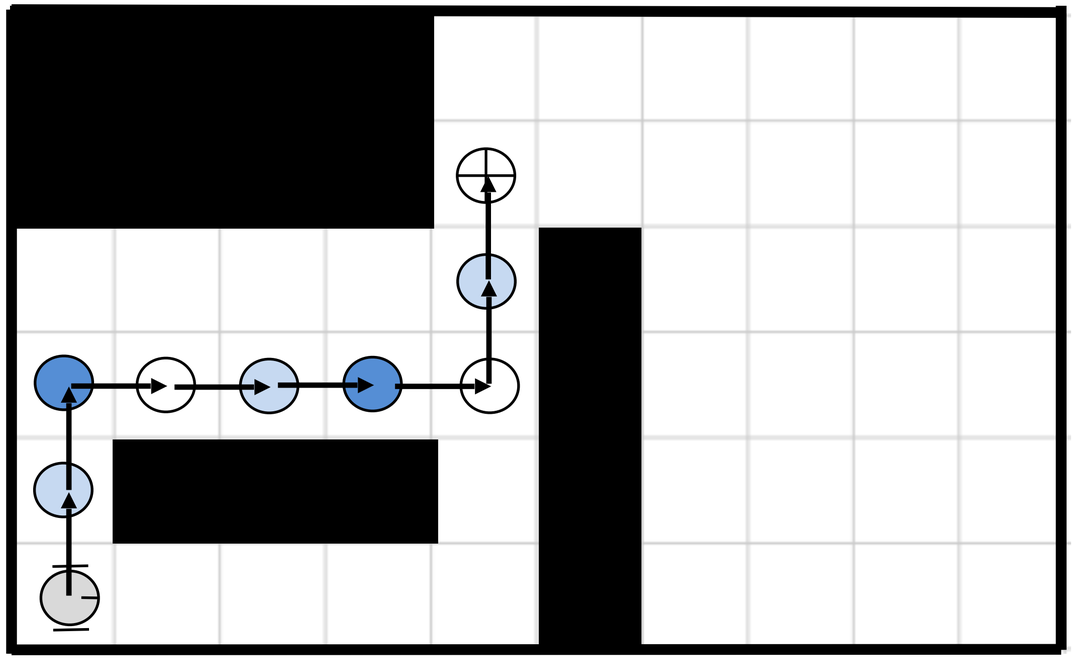




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Robot Motion & Planning





- **A* Characteristics**
  - Speed
    - $O(b^d)$ run time, **but Fastest BFS!**
  - Completeness
    - Yes
  - Optimality
    - Yes (wrt resolution)
  - Feasibility
    - Not guaranteed

